

3.3.1 Surface area to volume ratio

Content	Opportunities for skills development
The relationship between the size of an organism or structure and its surface area to volume ratio. Changes to body shape and the development of systems in larger organisms as adaptations that facilitate exchange as this ratio reduces. Students should be able to appreciate the relationship between surface area to volume ratio and metabolic rate.	PS 1.1 Students could use agar blocks containing indicator to determine the effect of surface area to volume ratio and concentration gradient on the diffusion of an acid or alkali. MS 4.1 Students could be given the dimensions of cells with different shapes from which to calculate the surface area to volume ratios of these cells.